

Ontology-Driven Clustering for Extracting Legal Reasoning Graphs

Anonymous ACL submission

Abstract

Extracting structured reasoning from long court judgments is a persistent challenge in legal NLP: large language models can identify individual legal concepts but struggle to consistently link them across thousands of tokens, producing graphs that are incomplete, structurally invalid, or non-reproducible. We present LEXCGRAPH, a pipeline that decomposes long-document graph extraction into a set of local inference problems by clustering extracted nodes around canonical legal concepts defined in a compiled ontology. Within each cluster, an LLM infers edges under ontology-provided constraints (definitions, AND/OR requirements, defeaters, relation whitelists); all cross-cluster structure is constructed deterministically. On 2,517 Indian Supreme Court cases from IL-TUR, ontology-normalized annotations outperform BM25 by +0.100 nDCG@10 ($p < 0.05$) and all dense retrieval baselines. Graph representations improve zero-shot outcome prediction by +11.1pp on Grok-4.1 and +11.1pp on Claude Sonnet 4.5 (McNemar $p < 0.05$), with substantially improved calibration (ECE: 0.109→0.026). Crucially, structured nodes *without* ontology-constrained edges perform *worse* than raw text, demonstrating that naive structuring destroys implicit textual signals—only ontology-constrained relational structure recovers and exceeds them. A leakage audit confirms comparable information access between representations (0.233 vs. 0.225 signals per 1K characters), and the consistent performance ordering across representations rules out LLM memorization. An ontology completeness ablation (11 vs. 54 concepts) reveals that enrichment dramatically improves graph quality (gold tier: 77%→90%, pseudo-cluster nodes: 93% reduction) while prediction accuracy remains unchanged—indicating that core doctrines saturate outcome-predictive signal while additional concepts improve structural fidelity for downstream applications. The

AND/OR justification structure enables counterfactual impact analysis: across 15,276 justification sets, 77.5% of holdings are structurally fragile, and removing the single most critical concept eliminates one-third of a case’s justifications on average. Cross-jurisdictional evaluation on 297 ECHR cases shows graph representations improve macro-F1 by +7.2pp on GPT-5.4, with minority-class recall gains of +17.1pp; calibration improvements replicate across jurisdictions. We release the ontology, extraction code, and evaluation scripts.¹

1 Introduction

Court judgments are long, multi-issue documents whose reasoning is distributed across factual findings, statutory interpretation, doctrine-specific tests, and citations to precedent. Capturing this reasoning in structured form is essential for legal case retrieval, explainability, and corpus-level legal analytics—yet the very properties that make court reasoning rich also make it difficult to extract: legally relevant spans are separated by thousands of tokens, doctrinal tests involve conjunctive and disjunctive requirements that must be tracked jointly, and the same abstract legal concept manifests differently across jurisdictions.

Large language models can extract individual nuggets of information from legal text, but they struggle to consistently link distant items across long inputs. When tasked with end-to-end graph extraction from a 15,000-token judgment, LLMs frequently produce graphs that are incomplete, structurally invalid, or non-reproducible. These failures are not primarily linguistic—the models understand the text—but combinatorial: the space of possible edges grows quadratically with the number of extracted nodes, and without structural constraints, the model must navigate this space

¹Code: <https://anonymous.4open.science/r/lexcgraph>

LexCGRAPH Pipeline: Ontology-Driven Legal Reasoning Graph Extraction

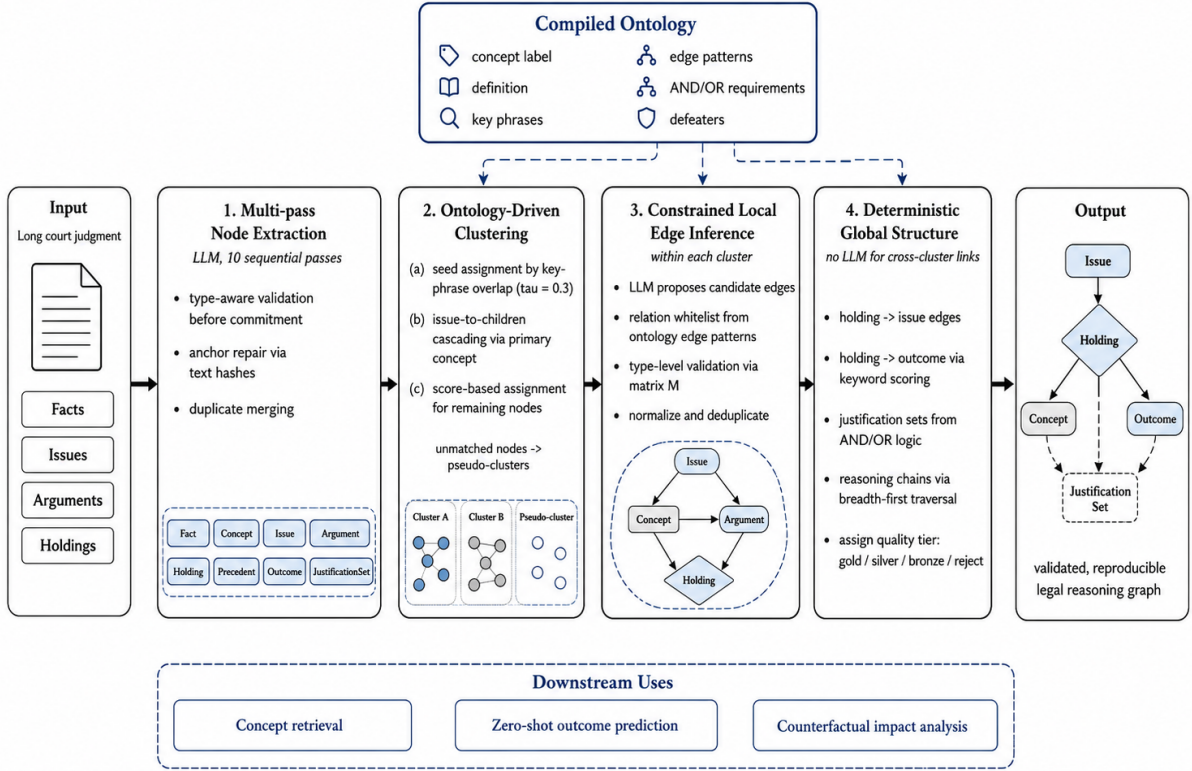


Figure 1: Overview of the LEXCGRAPH pipeline. A compiled ontology provides concept definitions, key phrases, edge patterns, AND/OR requirements, and defeaters. The four-stage pipeline decomposes long-document graph extraction into local inference problems, producing validated legal reasoning graphs for concept retrieval, zero-shot outcome prediction, and counterfactual impact analysis.

with only implicit knowledge of legal reasoning patterns.

We propose LEXCGRAPH, an ontology-driven approach that addresses this problem by decomposing long-document graph extraction into a set of *local* inference problems. The key mechanism is **ontology-driven clustering**: extracted nodes are assigned to clusters defined by canonical legal concepts in a compiled ontology, and edge inference is restricted to within-cluster pairs using ontology-provided constraints. All cross-cluster structure is constructed deterministically. This design yields three properties: (1) *reproducibility*—the same input always produces the same graph; (2) *structural validity*—every edge satisfies type-level constraints; and (3) *counterfactual traceability*—the AND/OR justification structure enables deterministic impact analysis when a concept node is removed.

Our evaluation reveals several findings with implications beyond legal NLP. First, ontology-constrained graphs substantially outperform raw

text on zero-shot prediction (+11.1pp) and concept retrieval (+0.100 nDCG@10). Second, and more surprisingly, structured nodes *without* ontology-constrained edges perform *worse* than raw text—demonstrating that naive structuring destroys implicit textual signals (discourse cues, paragraph structure, narrative flow) without replacing them with relational structure. Third, an ontology completeness ablation shows that expanding from 11 to 54 concepts dramatically improves graph structural quality (gold tier: 77% $\rightarrow$ 90%) without changing prediction accuracy, indicating that core doctrinal concepts saturate outcome-predictive signal. Fourth, a facts-only experiment confirms that the pipeline’s value derives from structuring *judicial reasoning*—the analytical connections between doctrines, precedents, and holdings—not from reorganizing factual content: graphs extracted from case facts alone provide no prediction advantage. These findings collectively establish that the graph’s value lies in structured representation of reasoning, not in information advantage

or superficial reorganization.

Contributions.

1. **Ontology-driven clustering for legal graph extraction:** a method that decomposes long-document graph construction into local inference problems scoped by ontology-defined concept clusters, producing reproducible, structurally validated graphs with deterministic cross-cluster structure (§3).
2. **Multi-dimensional evaluation:** prediction, retrieval, calibration, ontology completeness ablation, facts-only control, leakage audit, and counterfactual structural analysis—establishing both the value and the boundaries of graph-based legal reasoning (§5).
3. **Counterfactual impact analysis:** a domain-grounded explainability mechanism enabled by AND/OR justification logic, with corpus-level structural statistics across 15,276 justification sets (§6).
4. **Resources:** a compiled legal ontology (54 fully-specified concepts), extraction code, and evaluation scripts.

2 Related Work

Legal NLP and judgment prediction. LexGLUE (Chalkidis et al., 2022) and Legal-Bench (Guha et al., 2023) standardize legal NLP evaluation; IL-TUR extends evaluation to Indian law with the CJPE task (Joshi et al., 2024), derived from ILDC (Malik et al., 2021). For ECHR prediction, Chalkidis et al. (2019) established neural baselines; COLIEE (Rabelo et al., 2022) benchmarks statute retrieval. Structured prediction approaches model subtask dependencies as DAGs (Zhong et al., 2018), apply relational learning (Dong and Niu, 2021), and survey the state of the art (Feng et al., 2022). Our work differs in extracting the reasoning structure itself as a first-class output, rather than using implicit structure to improve a classifier.

Legal knowledge graphs and ontologies. The LKIF Core ontology (Hoekstra et al., 2007) provides foundational concepts for legal knowledge interchange; Küçük and Can (2025) survey datasets, benchmarks, and ontologies for computational law. Sarika et al. (2022) build a knowledge graph from Indian judgments using rule-based NER; Dhani et al. (2024) combine legal KGs with GNNs for case similarity. Our approach

uses the ontology not merely as a schema but as a *constraint* on the extraction process—scoping which edges the LLM may propose and enforcing structural validity at each step.

Structure-aware legal retrieval. Li et al. (2023) pre-train a structure-aware model for legal case retrieval, Ma et al. (2023) show structural information improves retrieval, and Tang et al. (2024) use inductive graph learning. Trippas et al. (2025) study reproducibility of graph-based retrieval—a problem our deterministic cross-cluster structure directly addresses.

Argumentation mining and graph reasoning. Toulmin (1958) provides the foundational claim/grounds/warrant abstraction; Mochales Palau and Moens (2011) and Wyner et al. (2010) developed automatic argumentation extraction from legal texts. Graph-of-Thoughts (Besta et al., 2024) models LLM reasoning as graph traversal but enforces no domain-specific constraints. Unlike graph counterfactual explainers that perturb edges to flip black-box predictions (Tan et al., 2022), our AND/OR justification structure enables domain-grounded counterfactual analysis that mirrors legal reasoning practice (§6).

3 Method

LEXCGRAPH extracts a legal reasoning graph from a court decision in four stages: (1) multi-pass node extraction, (2) ontology-driven clustering, (3) constrained local edge inference, and (4) deterministic global structure construction.

3.1 Graph Schema

Each decision is represented as a typed, anchored graph $G = (V, E, A)$. The schema defines eight node types (Fact, Concept, Issue, Argument, Holding, Precedent, Outcome, Justification-Set) and 32 edge relations derived from ontology edge patterns and legal argumentation literature (Toulmin, 1958; Mochales Palau and Moens, 2011). A type-level validation matrix $M \in \{0, 1\}^{|T| \times |T| \times |R|}$ specifies permitted relations between each pair of node types. Edges are normalized through an alias mapping layer (~ 200 aliases $\rightarrow 32$ canonical relations) and deterministically deduplicated.

3.2 Compiled Ontology

The compiled ontology serves dual roles: defining cluster keys and providing extraction con-

straints. Each concept includes: (1) label and definition, (2) key phrases for node-to-cluster assignment, (3) typical edge patterns constraining intra-cluster inference, (4) AND/OR requirements encoding conjunctive/disjunctive test structures, and (5) defeaters specifying blocking conditions.

The Indian ontology contains 54 fully-specified concepts: 11 constitutional doctrines (e.g., Basic Structure, Maneka Gandhi proportionality framework, Wednesbury Unreasonableness) with complete AND/OR reasoning structures, and 43 offense, procedural, and evidence concepts (e.g., Murder, Regular Bail, Electronic Evidence, Dying Declaration) with legal elements, defenses, establishing cases, and key phrases. All concepts were verified against primary legal sources with landmark case citations validated by independent audit (§8). Appendix A provides representative examples spanning each concept category and jurisdiction.

Concrete example: proportionality. The Maneka Gandhi proportionality framework is encoded as a six-part AND test: procedure by law must exist; it must be “right, just and fair”; not “arbitrary, fanciful or oppressive”; must satisfy Article 14 (non-arbitrariness); Article 19 (reasonableness); and conform to natural justice. Defeaters include express statutory exclusion and emergency situations. Key phrases include “just, fair and reasonable” and “Golden Triangle.” This structure scopes LLM edge inference: within a proportionality cluster, the LLM may only propose edges from the concept’s relation whitelist, and each proposed edge is validated against the type-level matrix M .

3.3 Extraction Pipeline

Stage 1: Multi-pass node extraction. Ten sequential passes with Grok-4.1, each validated before commitment: the type-aware matrix rejects invalid node types, anchors are repaired via text hashes, and duplicates are merged.

Stage 2: Ontology-driven clustering. Nodes are assigned to concept clusters via three phases: (1) seed assignment from key-phrase overlap ($\tau = 0.3$), (2) cascading of Issue nodes to children via primary-concept fields, and (3) score-based assignment for remaining nodes. Unmatched nodes enter pseudo-clusters with conservative constraints. Each cluster is summarized into a structured context record containing concept

definition, AND/OR requirements, defeaters, and candidate node mentions.

Stage 3: Constrained local edge inference.

Within each cluster, an LLM proposes edges under two constraints: a relation whitelist from the ontology’s edge patterns, and type-level validation via matrix M . Proposed edges are normalized and deduplicated.

Stage 4: Deterministic global structure.

All cross-cluster structure is constructed without LLM involvement: `resolves` edges from holdings to issues, `determines` edges from holdings to outcomes via keyword scoring, justification sets compiling intra-cluster evidence under AND/OR logic, and reasoning chains via breadth-first traversal. Each graph receives a quality tier (gold/silver/bronze/reject).

4 Experimental Setup

4.1 Dataset and Task

We evaluate on 2,517 IL-TUR CJPE cases (Joshi et al., 2024) (838 accepted, 1,679 rejected; stratified, seed=42). For zero-shot outcome prediction, we compare two input representations, both identically blinded: (i) *Raw*: decision text with outcome-revealing content removed via regex filtering; and (ii) *Graph*: linearized reasoning graph with outcome-bearing fields withheld.

ECHR cross-jurisdiction set. For cross-jurisdictional evaluation we use 297 full-judgment ECHR merits decisions scraped from HUDOC (Turkey-respondent cases) (Chalkidis et al., 2019): 262 violation, 35 no-violation (88% imbalance). A separately compiled 55-concept ECHR ontology covers Convention articles, doctrinal tests (margin of appreciation, proportionality, prescribed-by-law), and procedural principles (exhaustion of domestic remedies, victim status, four-month rule). Extraction is performed by GPT-4.1 mini (OpenAI, 2025) rather than a frontier model, deliberately testing pipeline robustness under weaker extraction. The blinding protocol is identical to IL-TUR (outcome-bearing graph fields withheld; regex-based outcome removal from raw text, plus the HUDOC metadata banner stripped), and the raw-text character budget is set to $\approx 1.05 \times$ the average graph summary length to match information access at the prompt level.

Table 1: Zero-shot CJPE outcome prediction on 2,517 IL-TUR cases. Graph inputs outperform raw text on both models across all metrics.

Model	Input	Acc.	F1	ECE↓	Brier↓
Grok-4.1	Raw	.742	.722	.109	.195
	Graph	.853	.839	.026	.120
Sonnet 4.5	Raw	.663	.631	.091	.228
	Graph	.774	.754	.010	.174

4.2 Models and Metrics

For IL-TUR we evaluate on two architecturally distinct frontier models: Grok-4.1 and Claude Sonnet 4.5. Since Grok also serves as the extraction model, Sonnet provides an independent cross-model control. For the ECHR set we use GPT-4.1 mini (OpenAI, 2025)—which is both extractor and a non-frontier predictor—and GPT-5.4 (OpenAI, 2026) as a frontier predictor on the same graphs. Using a non-frontier extractor stress-tests whether the ontology constraints can compensate for weaker upstream extraction. We measure accuracy, macro-F1, calibration (ECE, Brier), and use McNemar’s test with bootstrap 95% CIs. For retrieval, we report nDCG@10, MAP, and P@10 with paired t -tests. Our IL-TUR evaluation across 10,068 model runs produced zero parse or API errors.

5 Results

5.1 Outcome Prediction

Table 1 shows that graphs consistently outperform raw text. Accuracy improves by +11.1pp on Grok and +11.1pp on Sonnet (McNemar $\chi^2=119.97$ and 92.34; $p < 0.05$), with bootstrap 95% CIs excluding zero. Calibration improves markedly: ECE drops from 0.109→0.026 on Grok, with raw text systematically overconfident in the 0.70–0.90 range while graph inputs track the reliability diagonal closely.

5.2 Structure vs. Structured Nodes

Table 2 reveals a striking finding: structured nodes without ontology-constrained edges perform *worse* than raw text on both models. On Grok, structured nodes achieve 0.718 accuracy versus 0.739 for raw text. This demonstrates that merely extracting and labeling legal concepts can *destroy* implicit textual signals—discourse cues, paragraph structure, narrative flow—without replacing them with relational structure. Only when

Table 2: Ablation: Full Graph vs. Structured Nodes (same nodes, no edges) vs. Raw text. Structured nodes without edges perform *worse* than raw text. †McNemar $p < 0.05$.

Model	Metric	Full Graph	Struct. Nodes	Raw
Grok-4.1	Acc.	0.864 †	0.718	0.739
	Brier↓	0.113	0.211	0.196
Sonnet 4.5	Acc.	0.771 †	0.683	0.663
	Brier↓	0.175	0.219	0.228

Table 3: Ontology completeness ablation. Expanding from 11 to 54 concepts dramatically improves graph quality while prediction accuracy remains unchanged.

Metric	11 concepts	54 concepts
Prediction accuracy	0.855	0.853
Gold tier graphs	77.1%	90.3%
UNLISTED nodes/graph	47.4	3.2
Edges/graph	108.8	116.0
Unique concept IDs	14,064	1,634

ontology-constrained edges, justification sets, and reasoning chains are added does the graph representation substantially exceed both baselines. The consistent performance ordering across representations (full graph > raw text > structured nodes) using identical models and cases rules out LLM memorization as an explanation: if models recognized cases from pretraining, performance would be independent of input representation.

5.3 Ontology Completeness Ablation

To assess the marginal value of ontology investment, we expanded the Indian ontology from 11 fully-specified doctrinal concepts to 54 concepts (adding 43 offense, procedural, and evidence concepts with complete legal elements, defenses, and key phrases) and re-ran the full pipeline on the same 2,517 cases.

Table 3 reveals a dissociation between graph quality and prediction accuracy. Graph structural quality improved dramatically: gold-tier graphs increased from 77.1% to 90.3%, pseudo-cluster (UNLISTED) node assignments dropped by 93% (47.4→3.2 per graph), and 14,064 ad-hoc pseudo-concept IDs collapsed into 1,634 canonical ontology concepts. However, prediction accuracy was statistically unchanged (0.855 vs. 0.853, $p > 0.05$).

This indicates that the 11 core doctrinal concepts—proportionality, natural justice, reasonable classification, Wednesbury unreasonableness,

and others—capture the reasoning structure predictive of case outcomes. The additional 43 offense-level and procedural concepts improve structural completeness for applications requiring fine-grained graph fidelity (retrieval, explainability, counterfactual analysis) without contributing additional predictive signal for binary outcomes.

5.4 Facts-Only Control

A natural question is whether the pipeline’s value derives from structuring judicial reasoning or merely from reorganizing factual content. To test this, we extracted graphs from only the facts section of 50 cases (the same input available to supervised ILDC baselines) and compared prediction accuracy across three conditions: raw facts (64%), facts-only graph (62%), and full graph (90%). The facts-only graph provides no prediction advantage, confirming that the pipeline’s value derives from structuring the court’s analytical reasoning—the connections between doctrines, precedents, and holdings—not from reorganizing facts. This situates LEXCGRAPH as a reasoning structure extraction tool rather than a fact enhancement method.

5.5 Fairness Controls

Information leakage audit. Both representations undergo equivalent outcome blinding. An audit of 30 cases measuring outcome-signaling patterns found comparable leakage density: raw text at 0.233 signals per 1K characters versus graph at 0.225 (1.0 $\times$ ratio). The graph eliminates discourse-level judicial signals (e.g., “in our opinion”: 3 instances in raw, 0 in graph) while inheriting comparable treatment-label leakage from precedent nodes. The accuracy gap is therefore attributable to structural organization, not differential information access.

Memorization defense. The consistent ordering (full graph > raw text > structured nodes) using identical models and cases rules out LLM memorization. No memorization hypothesis explains why removing edges from an extracted graph would make the LLM perform *worse* than raw text. Only a representation-quality hypothesis explains this pattern.

5.6 Concept Retrieval

Table 4 shows that ConceptSet (ontology-normalized graph annotations) substantially outperforms all baselines including BM25

Table 4: Concept retrieval (48 queries, 2,517 cases). $\dagger p < 0.05$ vs. BM25.

Method	Type	nDCG@10	MAP	P@10
Legal-BERT	Bi-enc.	0.189	0.082	0.175
InLegalBERT	Bi-enc.	0.235	0.098	0.215
E5-large	Bi-enc.	0.346	0.160	0.323
BGE-large	Bi-enc.	0.350	0.158	0.333
ColBERTv2	Late-int.	0.425	0.191	0.390
BM25	Sparse	0.732	0.461	0.702
ConceptSet [†]	Structured	0.832	0.510	0.831

Table 5: Cross-jurisdictional ECHR evaluation (297 Turkey-respondent HUDOC cases, 88% violation). Extraction: GPT-4.1 mini. Graph representations improve macro-F1 at both predictor scales; the advantage grows with model capability (+4.5pp on mini, +7.2pp on GPT-5.4). Neither accuracy gap reaches McNemar significance due to class-imbalance-compressed headroom (majority baseline: 88%).

Predictor	Input	Acc.	Macro-F1	ECE↓	Brier↓
GPT-4.1 mini	Raw	.906	.670	.082	.092
	Graph	.906	.715	.063	.085
GPT-5.4	Raw	.926	.751	.037	.063
	Graph	.939	.823	.035	.054

(+0.100 nDCG@10, +0.129 P@10; paired t -test, $p=0.016$). A clear four-tier ranking emerges: domain-adapted legal models perform worst, general-purpose bi-encoders improve, ColBERTv2 partially recovers lexical signal, and BM25 dominates all dense baselines—reflecting concept-level legal retrieval’s dependence on exact term matching. ConceptSet additionally identifies semantically relevant cases lacking explicit surface markers (Jaccard overlap between regex-derived and annotation-derived relevance: 0.43).

5.7 Cross-Jurisdictional Evaluation

Table 5 shows that on ECHR, graph representations beat raw text across accuracy, macro-F1, ECE, and Brier on GPT-5.4, and on macro-F1 and calibration on GPT-4.1 mini. Macro-F1 is the appropriate primary metric given the 88% class imbalance, and the +7.2pp graph advantage on GPT-5.4 is driven entirely by minority-class recall: graphs correctly classify 54.3% of no-violation cases versus 37.1% for raw (+17.1pp), while both methods correctly identify >99% of violations. The accuracy gap of +1.4pp does not reach McNemar significance ($\chi^2=0.45$) because the 88% majority baseline compresses accuracy headroom—

Table 6: Counterfactual structural statistics across 2,446 graphs (15,276 justification sets, 14,483 holdings with justification).

Metric	Value
Total justification sets	15,276
AND logic	11,877 (77.7%)
OR logic	3,399 (22.3%)
Avg. evidence edges per JS	4.8
Fragile holdings (AND only)	11,220 (77.5%)
Robust holdings (≥ 1 OR path)	2,606 (18.0%)
Mixed holdings	657 (4.5%)
Mean fragility score	0.344
Cases with fragility > 0.5	324 (13.2%)

the same +17.1pp minority-class win produces only ≈ 2 pp of overall accuracy at this base rate.

Capability scaling. The graph advantage grows with predictor capability: macro-F1 +4.5pp on GPT-4.1 mini versus +7.2pp on GPT-5.4, and no-violation recall rises from +11.4pp (mini) to +17.1pp (GPT-5.4). This is consistent with the §5 finding that the graph’s value lies in relational structure requiring active inference: stronger reasoners more effectively exploit the conjunctive/disjunctive justification structure that distinguishes a no-violation finding (e.g., margin-of-appreciation deference, prescribed-by-law satisfied, legitimate aim served) from a violation finding.

Extraction robustness. Graphs were extracted by GPT-4.1 mini, a non-frontier model, with 80% gold+silver quality tier (vs. 90% gold on IL-TUR with Grok-4.1). Despite this lower extraction quality, the prediction advantage persists at both predictor scales, indicating that ontology-provided edge whitelists and type-level validation compensate for weaker extraction. This evidence—together with the calibration improvement replicating across both jurisdictions—supports the claim that the pipeline’s benefits derive from the ontology-driven structural constraints, not from frontier-model extraction.

6 Counterfactual Impact Analysis

The AND/OR justification structure enables counterfactual impact analysis: given any concept node, the system deterministically traces which holdings lose support upon its removal.

Table 6 presents corpus-level statistics. 77.5% of holdings are supported exclusively by conjunc-

tive (AND) justification sets, meaning removal of *any* required concept eliminates the holding’s justification. The mean fragility score of 0.344 indicates that removing the single most critical concept eliminates approximately one-third of a case’s holdings—consistent with the multi-issue nature of appellate decisions. The most frequently load-bearing concepts are Article 14 (146 cases) and Natural Justice (102 cases), aligning with their prevalence in Indian constitutional litigation.

Doctrinal predictive asymmetry. Individual doctrines vary substantially in outcome directionality. When serving as the structurally critical concept, Reasonable Classification cases are rejected 87.9% of the time (reflecting the twin test’s deference to the legislature), while Natural Justice cases show near-equal outcome splits (50.8%), indicating that outcomes depend on which prongs of the multi-part test are satisfied rather than on doctrinal presence alone. This asymmetry suggests that the predictive value of graphs derives from capturing *which* elements of doctrinal tests are satisfied, not merely from identifying *which* doctrines appear.

7 Discussion

Why does clustering help? Long judgments contain legally relevant spans separated by thousands of tokens. Clustering around canonical concepts reduces edge inference from a quadratic all-pairs problem to a set of local problems. The deterministic cross-cluster structure prevents LLM “hallucinated” long-range edges and ensures reproducibility.

The graph as reasoning structure, not information advantage. Our results establish that the graph’s value derives specifically from structuring judicial reasoning. Three findings converge: (1) facts-only graphs provide no advantage (62% vs. 64% raw facts), confirming the signal comes from the court’s analytical sections; (2) the leakage audit shows comparable information access ($1.0\times$ density ratio); and (3) the ablation shows that structured nodes without edges perform *worse* than raw text, ruling out information advantage as an explanation. The graph organizes reasoning, not information.

When to invest in ontology expansion. The ontology completeness ablation provides practical guidance: 11 well-specified doctrinal concepts are sufficient for prediction, but 54 concepts are

needed for high-quality graphs (90% gold) suitable for retrieval, explainability, and counterfactual analysis. This suggests a two-stage adoption path: start with core doctrines for immediate prediction benefits, then expand for richer downstream applications.

Ontology compilation cost. The 54-concept Indian ontology was compiled in approximately 16 hours of domain expert effort, with 43 offense/procedural/evidence concepts enriched via LLM-assisted generation followed by independent legal verification (93% of concepts passed verification; 2 errors corrected). This hybrid approach—LLM generation with expert audit—suggests a scalable methodology for adapting to new jurisdictions.

What does and does not generalize across jurisdictions. The ECHR evaluation (§5.7) shows the pipeline ports beyond Indian law using a separately compiled 55-concept ontology built under the same schema. Two findings emerge. (1) *Calibration improvement is the most robust cross-jurisdictional finding:* graph reduces both ECE and Brier at every model scale and on both datasets, even when accuracy gaps are small. (2) *Effect size on accuracy depends jointly on predictor capability and dataset class balance:* at 88% class imbalance the majority baseline absorbs much of the headroom, so the same minority-class wins that drive +11.1pp on balanced IL-TUR produce only +1.4pp on ECHR. The macro-F1 view (+7.2pp on GPT-5.4) is a fairer summary in imbalanced settings.

8 Limitations

Ontology coverage. Out-of-ontology concepts fall back to pseudo-clusters with conservative constraints. While the 54-concept ontology reduced pseudo-cluster assignments by 93%, novel or emerging doctrines may be under-linked.

Information access. Our evaluation uses full judgment text with outcome blinding, not facts alone. The pipeline’s value derives from structuring judicial reasoning (confirmed by the facts-only control), which requires the court’s analytical sections as input. This limits applicability to post-decision analysis rather than pre-decision prediction.

Concept matching. Node-to-cluster assignment uses keyword overlap rather than learned

embeddings, prioritizing reproducibility over semantic precision.

Relevance judgments. Regex-based relevance labels for retrieval evaluation may under-count semantically relevant cases and over-count passing mentions.

Proprietary models. Our IL-TUR results use proprietary frontier models (Grok-4.1 for extraction, Grok and Sonnet for evaluation). The ECHR evaluation (§5.7) partially addresses this concern by extracting with GPT-4.1 mini (non-frontier) and reporting predictions on both that model and frontier GPT-5.4, demonstrating the pipeline’s benefits persist with weaker extractors. Fully open-weights reproducibility—e.g., extraction with a Llama- or Qwen-class model—remains future work.

Ontology verification. The 43 enriched offense/procedural concepts were generated with LLM assistance and verified by independent audit. Of 43 concepts, 22 passed cleanly, 6 had minor imprecisions, 2 had errors (corrected), and 13 contained case citations that could not be independently verified. Verification by a practicing Indian legal expert would strengthen confidence.

9 Conclusion

We introduced LEXCGRAPH, an ontology-driven pipeline for extracting legal reasoning graphs. By localizing LLM inference to ontology-defined concept clusters and constructing all global structure deterministically, we obtain reproducible, validated graphs that enable domain-grounded counterfactual impact analysis. On IL-TUR CJPE, graphs substantially improve zero-shot prediction and calibration, with ablations confirming that ontology-constrained relational structure—not clean node extraction, not information advantage, not LLM memorization—drives the gains. An ontology completeness ablation reveals that core doctrines saturate predictive signal while broader coverage improves structural fidelity, providing practitioners actionable guidance on ontology investment. The AND/OR justification structure enables counterfactual analysis at corpus scale, with 77.5% of holdings structurally fragile under concept removal.

The pipeline generalizes beyond legal NLP to any domain with structured diagnostic criteria where long documents require ontology-constrained decomposition for reliable

extraction—including clinical reasoning with diagnostic ontologies (DSM-5, ICD-11) and regulatory compliance with statutory requirement structures.

Ethics Statement

All data used in this work are publicly available court decisions from the IL-TUR corpus (Joshi et al., 2024) and ECHR cases (Chalkidis et al., 2019). The dataset contains no personal or sensitive information beyond what appears in the published judgments themselves, which are matters of public record.

Our system performs *post-decision* structural analysis of judgments that have already been decided and published. We do not deploy outcome predictions on pending or real-world cases; the counterfactual analysis is restricted to retrospective examination of decided judgments. We therefore see no direct risk of automated judicial decision-making arising from this work.

Limitations of the ontology verification process—including reliance on LLM-assisted generation for 43 of the 54 concepts and incomplete independent expert validation—are discussed in §8.

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A Ontology Examples

We present representative examples from each concept category and jurisdiction to illustrate the ontology’s structure. Each concept encodes the legal test as AND/OR requirements, blocking conditions as defeaters, and surface-level triggers as key phrases. The full ontologies are released with our code.

A.1 Indian Concepts

Maneka Gandhi Framework (Doctrine; Golden Triangle of Articles 14, 19, 21). *Definition:* State action affecting Article 21 must satisfy a six-part substantive test interlinking the Article 14, 19, and 21 guarantees, beyond the bare procedure-established-by-law requirement.

- AND Procedure established by law exists
- AND Procedure must be “right, just and fair”
- AND Procedure must not be “arbitrary, fanciful or oppressive”
- AND Must satisfy Article 14 (non-arbitrariness)
- AND Must satisfy Article 19 (reasonableness if applicable)
- AND Must conform to principles of natural justice

Defeaters: express statutory exclusion of natural justice; emergency situations; clear legislative intent excluding procedural requirements.

Key phrases: “just, fair and reasonable”; “right and just and fair”; “not arbitrary, fanciful or oppressive”; “Golden Triangle”.

Establishing cases: Maneka Gandhi v. UOI (1978); R.C. Cooper v. UOI (1970).

Edge pattern: (State_Action) $\xrightarrow{\text{must satisfy}}$ (Maneka_Gandhi_Test) $\xrightarrow{\text{if failed}}$ (Unconstitutional).

Murder (Offense; BNS §103 / IPC §302). *Definition:* Causing the death of a human being with the mens rea specified in IPC 300 / BNS 101 and not falling within any of the five exceptions.

Actus reus: act causing death of a human being
Mens rea (any one of four IPC 300 / BNS 101 alternatives): intention to cause death; *or* intention to cause bodily injury known to be likely to cause death; *or* intention to cause injury sufficient in the ordinary course of nature to cause death; *or* knowledge that the act is so imminently dangerous that it must in all probability cause death

- AND Death actually results from the act
- AND Act does not fall within any of the five exceptions to IPC 300 / BNS 101

Defeaters: private defence (IPC 96–106 / BNS 34–44); insanity (IPC 84 / BNS 22); accident without criminal intent (IPC 80 / BNS 20); act of a child under 7 (IPC 82 / BNS 17); sudden and grave provocation (Exception 1); sudden fight without premeditation (Exception 4); consent of victim above 18 (Exception 5).

Key phrases: “committed murder”; “Section 302”; “BNS 103”; “intention to kill”; “culpable homicide amounting to murder”; “imminently dangerous act”.

Establishing cases: Virsa Singh v. State of Punjab (1958); Rajwant Singh v. State of Kerala (1966); State of AP v. R. Punnayya (1977).

Edge pattern: (Accused_Act) $\xrightarrow{\text{causes}}$ (Death_of_Victim) $\xrightarrow{\text{with}}$ (Mens_Rea) $\xrightarrow{\text{not within exceptions}}$ (Conviction_BNS103).

Anticipatory Bail (Procedural; BNSS §482 / CrPC §438). *Definition:* Pre-arrest direction for release on bail granted by the High Court or Court of Session to a person apprehending arrest on accusation of a non-bailable offense.

- AND Applicant has reason to believe arrest may be

843	made on a non-bailable accusation	signed by an unauthorized person.	894
844	AND Application filed before the High Court or Court	<i>Key phrases:</i> “electronic evidence”; “Section	895
845	of Session (CrPC 438 / BNSS 482)	65B”; “BSA 63”; “certificate”; “hash value”;	896
846	AND Court considers: nature/gravity of accusation;	“electronic record”; “Anvar P.V. compliance”;	897
847	applicant’s antecedents; flight risk; whether the	“prescribed format”.	898
848	accusation is to humiliate or injure	<i>Establishing cases:</i> Anvar P.V. v. P.K. Basheer	899
849	AND Court may impose conditions: availability	(2014); Arjun Panditrao Khotkar v. Kailash	900
850	for interrogation; no inducement/threat to wit-	Kushanrao Gorantyal (2020); Shafhi Mohammad	901
851	nesses; no leaving India without permission	v. State of HP (2018, overruled by Khotkar).	902
852	AND BNSS 482(4) carve-out: anticipatory bail does	<i>Edge pattern:</i> (Electronic_Record)	903
853	not apply to BNS 65 (rape of a minor under 16)	valid 65B/63 cert + hash → (Admissible)	904
854	or BNS 70(2) (gang rape of a minor)	court evaluates authenticity → (Evidence_Weighed).	905
855	<i>Defeaters:</i> accusation falls under the		
856	BNSS 482(4) statutory exclusion; prima facie		906
857	case requiring custodial interrogation; flight		
858	risk; likelihood of witness tampering; apprehen-	Article 3 — Prohibition of Torture (Absolute	907
859	sion of arrest not bona fide.	Right). <i>Definition:</i> Absolute prohibition on tor-	908
860	<i>Key phrases:</i> “anticipatory bail”; “Section 438”;	ture and inhuman or degrading treatment; non-	909
861	“BNSS 482”; “reason to believe arrest may be	derogable under Article 15§2; no balancing per-	910
862	made”; “pre-arrest bail”; “custodial interroga-	mitted.	911
863	tion”; “humiliate or injure”.	AND Treatment crosses the <i>minimum level of sever-</i>	912
864	<i>Establishing cases:</i> Gurbaksh Singh Sibbia v.	ity threshold (relative: duration, effects, victim	913
865	State of Punjab (1980); Sushila Aggarwal v. State	characteristics)	914
866	(NCT of Delhi) (2020); Siddharam Satlingappa	AND Classification into a tier: torture (deliberate, pur-	915
867	Mhetre v. State of Maharashtra (2011).	posive) > inhuman treatment (intense suffering)	916
868	<i>Edge pattern:</i> (Apprehension_of_Arrest)	> degrading treatment (humiliation)	917
869	HC/Sessions → (Court_Evaluates_BNSS482)	<i>Defeaters:</i> none — the right is absolute. No	918
870	if no exclusion → (Anticipatory_Bail_Granted).	exceptions, no derogation, no balancing.	919
871	Electronic Evidence (Evidence; BSA §63 / IEA	<i>Key phrases:</i> “minimum level of severity”;	920
872	§65B). <i>Definition:</i> Admissibility of computer	“absolute prohibition”; “no derogation”; “non-	921
873	outputs and digital records requires a certificate	refoulement”; “substantial grounds for believing”;	922
874	authenticating the production process and the in-	“real risk”; “most fundamental values”.	923
875	tegrity of the data.	<i>Establishing cases:</i> Ireland v. UK (1978); Soering	924
876	AND Evidence is an electronic record (as defined in	v. UK (1989); Chahal v. UK (1996); Selmouni v.	925
877	the IT Act 2000)	France [GC] (1999); Saadi v. Italy [GC] (2008);	926
878	AND Accompanied by a certificate under IEA 65B(4)	M.S.S. v. Belgium and Greece [GC] (2011).	927
879	/ BSA 63(4) identifying the record, describing	<i>Edge pattern:</i> (Treatment) $\xrightarrow{\text{min severity?}}$	928
880	its production, and giving particulars of the de-	(Torture/Inhuman/Degrading) $\xrightarrow{\text{if met}}$ (Auto-	929
881	vice used	matic_Violation).	930
882	AND Under BSA 63: certificate in prescribed format;	Article 8 — Right to Respect for Private and	931
883	hash value of the record is <i>mandatory</i> ; signed	Family Life (Qualified Right). <i>Definition:</i> Pro-	932
884	by both the person in charge of the device <i>and</i>	tection of private life, family life, home, and cor-	933
885	an expert	respondence; restrictions are permitted only when	934
886	AND Computer was regularly used for lawful activi-	the four-step interference test under Article 8§2 is	935
887	ties during the relevant period; information was	satisfied.	936
888	fed in the ordinary course of business	AND State action constitutes an <i>interference</i> with pri-	937
889	<i>Defeaters:</i> no IEA 65B(4) / BSA 63(4) certificate	ate life, family life, home, or correspondence	938
890	(per Anvar P.V.); certificate fails prescribed format	AND Interference is <i>prescribed by law</i> (accessible and	939
891	or hash requirement; hash mismatch indicating	foreseeable)	940
892	tampering; broken chain of custody; computer	AND Interference pursues a <i>legitimate aim</i> from the	941
893	malfunction in the relevant period; certificate	Article 8§2 closed list	942

AND Interference is *necessary in a democratic society* (pressing social need plus proportionate response)

Defeaters: interference is in fact prescribed by law; pursues a legitimate aim; is necessary and proportionate; State acted within its margin of appreciation.

Key phrases: “private life”; “family life”; “home”; “correspondence”; “personal data”; “physical and psychological integrity”; “surveillance safeguards”.

Establishing cases: Dudgeon v. UK (1981); López Ostra v. Spain (1994); Von Hannover v. Germany (No. 2) [GC] (2012); S. and Marper v. UK (2008); Zakharov v. Russia (2015).

Edge pattern: (Interference_with_Art8_right) $\xrightarrow{\text{four-step test}}$ (Prescribed + Aim + Necessary) $\xrightarrow{\text{if fails}}$ (Violation).

A.3 Turkish Concepts

Right to Personal Liberty and Security (Art. 19, 1982 Constitution; Kişi Özgürlüğü ve Güvenliği). *Definition:* Constitutional protection against unlawful or arbitrary detention. The same ontology schema is adapted to Turkish criminal procedure (*Ceza Muhakemesi Kanunu*, CMK) and Anayasa Mahkemesi (AYM) bireysel başvuru jurisprudence; structurally parallel to ECHR Article 5.

AND *Kuvvetli suç şüphesi* — strong suspicion based on concrete evidence (CMK m.100)

AND At least one of: *kaçma şüphesi* (flight risk) **or** *delilleri karartma* (evidence tampering)

AND Maximum detention period under CMK m.102 not exceeded: 1.5 years (non-heavy penal cases), 5 years (heavy penal cases), 7 years (enumerated offenses including the Anti-Terror Law list)

AND Alternative measures (*adli kontrol*, CMK m.109) considered before detention

Defeaters: CMK m.100 detention requirements not met; CMK m.102 maximum period exceeded; no concrete evidence supporting strong suspicion; *adli kontrol* alternatives not considered; automatic detention without individualised assessment.

Key phrases: “kişi özgürlüğü”; “tutuklama”; “kuvvetli suç şüphesi”; “kaçma şüphesi”; “delilleri karartma”; “gözaltı”; “adli kontrol”; “tutukluluk süresi”.

Establishing cases: AYM, Mehmet Haberal [GK], B.No. 2012/849, 4/12/2013 (foundational

Art. 19 bireysel başvuru); AYM, Murat Narman, B.No. 2012/1137, 2/7/2013; AYM, Hikmet Kopar ve diğerleri [GK], B.No. 2014/14061, 8/4/2015 (detention lawfulness criteria).

Edge pattern: (Detention_Order) $\xrightarrow{\text{CMK m.100 test}}$ (Kuvvetli_Şüphe + Tutuklama_Nedeni) $\xrightarrow{\text{if not met}}$ (Art. 19_Violation).

B Quality Tier Distribution

Ontology	Gold	Silver	Bronze	Reject
11 concepts	77.1%	20.7%	1.7%	0.6%
54 concepts	90.3%	7.5%	1.8%	0.4%